Bilkent University
Department of Electrical and Electronics Engineering

EEE 431 - Digital Communications

Computational Assignment 1

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Introduction

In this assignment, we are asked to solve various quantization problems. Firstly, we are to obtain a desired PDF from Uniform(0,1). We will then calculate the SQNR for uniform and non-uniform quantization cases. We calculate the non-uniform quantizer using the Lloyd-Max Algorithm. We will calculate the PMFs of quantizer outputs for both cases.

For the second question, we will work with two images, foliage.tif and an image of our choosing. We will obtain the approximate PDFs for the pixel values. We will inspect what quantization noise looks like in the spatial and Fourier domains for images. We will then LPF the original and quantized images to compare them.

Question 1

Part-A

We wanna obtain a triangular PDF from a rectangular PDF. We can achieve this by adding two random variables with rectangular distributions. The resulting PDF will be their convolution, ie, a triangular function.

$$X_1, X_2 \sim Uniform(0,1),\ R = X_1 + X_2 - 1,\ f_R(r) = \Lambda(r)$$

Alternatively:

$$x = \text{sign}(2(U-0.5))\left(1 - \sqrt{|2(U-0.5)|}\right).$$

Part-B

We can see that the histograms converge to the PDF of X.

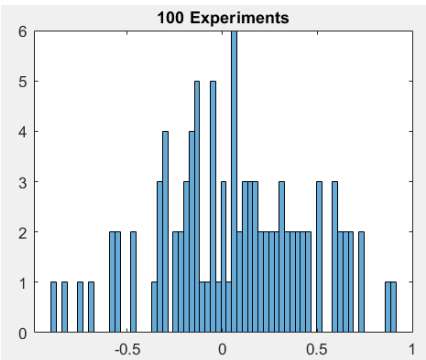

Figure 1.2.1: Histogram for 100 Experiment Outputs

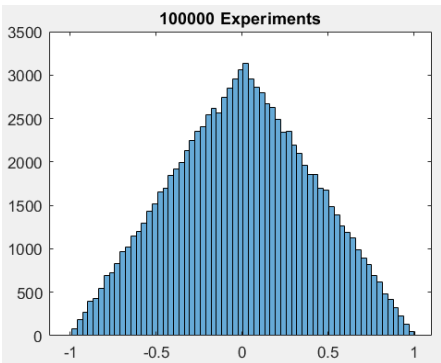

Figure 1.2.2: Histogram for 100000 Experiment Outputs

Part-C

We can calculate the SQNR for each case with the following calculations [1]:

$SQNR = 10\log_{10}(\frac{E[X^2]}{X_{max}^2}) + 6.02v + 4.8$	$SQNR = 10\log_{10}(\frac{E[X^2]}{X_{max}^2}) + 6.02v + 4.8$	$SQNR = 10\log_{10}(\frac{E[X^2]}{X_{max}^2}) + 6.02v + 4.8$
$E[X^2] = 2\int_0^1 (1-x)x^2 dx = \frac{1}{6},\ v = 3$	Similarly, $v = 4$	Similarly, $v = 6$
So SQNR = 15.07dB so MSE = 518.61e-5	So SQNR = 21.09dB so MSE = 129.67e-5	So SQNR = 33.13dB so MSE = 8.106e-5

Note that these MSEs are for a single experiment. The MSE values in the following parts are for 10^5 experime iş nts so they are scaled.

Part-D

Plotting the histograms of quantization outputs, we obtain the following results. We can see that the SQNR values are approximately the same, with some possible errors stemming from floating-point arithmetic errors [2] and potential errors during the approximation of the analytical formula. (Note that the MSEs are for 100000 outputs.)

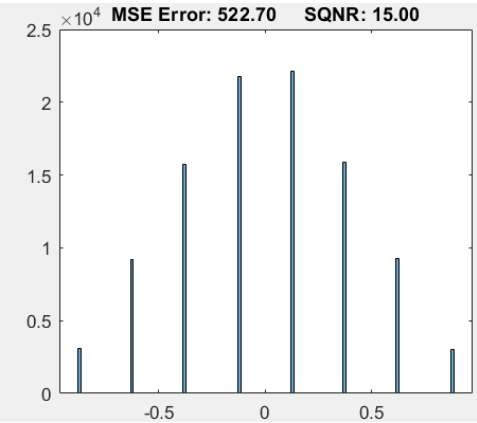

Figure 1.4.1: Quantization outputs for N=8

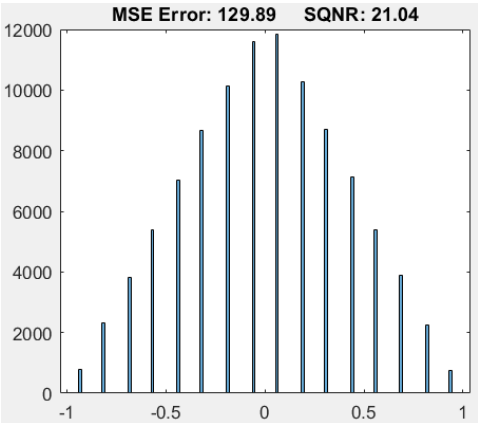

Figure 1.4.2: Quantization outputs for N=16

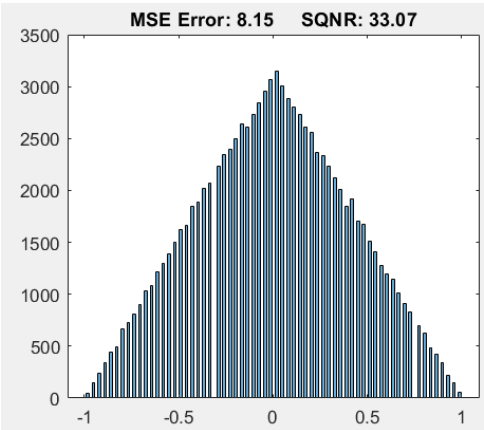

Figure 1.4.3: Quantization outputs for N=64

Part-E

We can model the quantizer as dividing the PDF into trapezoids. We can infer that the probability of each quantization level will be equal to the area of the trapezoid between its respective quantization boundaries. We can calculate the PMFs by calculating those areas. We can realise that the PMF is approaching the PDF

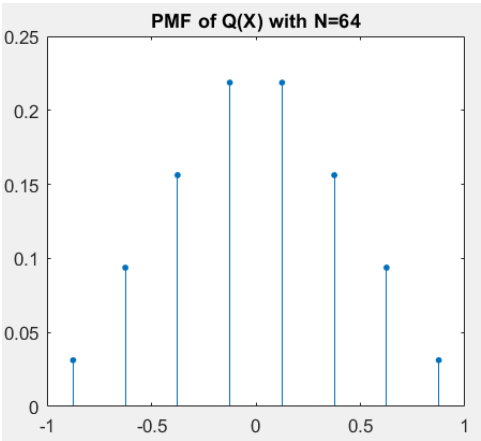

Figure 1.5.1: Quantization Output PMF for N=8

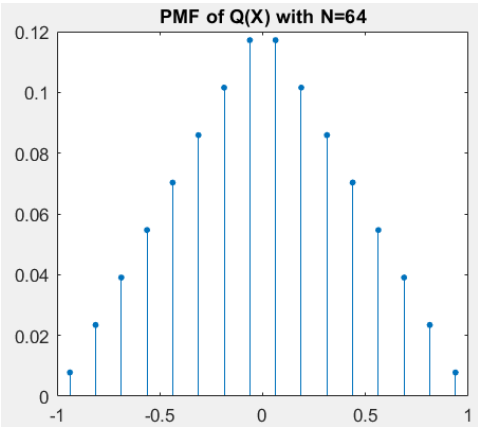

Figure 1.5.2: Quantization Output PMF for N=16

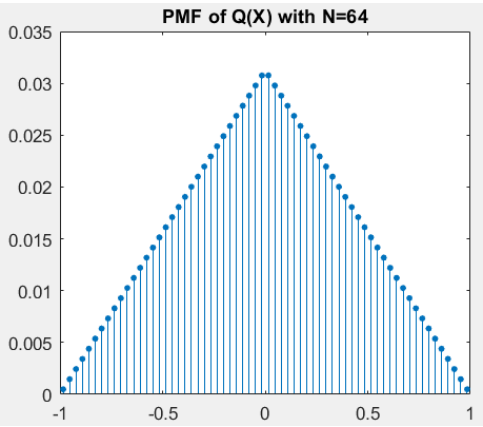

Figure 1.5.3: Quantization Output PMF for N=64

Part-F

The Lloyd-Max algorithm is an iterative algorithm used to find the optimal non-uniform quantizer for a given PDF [2]. For non-convex PDFs, it is guaranteed to find the optimal quantizer. Starting with arbitrary values, the algorithm consists of three steps:

1. Select each quantization level as the expectation in its respective interval.
2. Select quantization boundaries in the middle of two adjacent quantization levels.
3. If the convergence slows down to the desired level, stop. If not, go back to step 1.

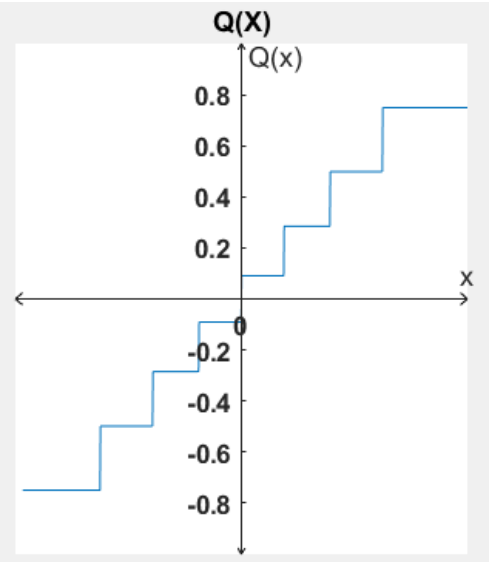

Figure 1.6.1: Non-Uniform Quantization function for N=8

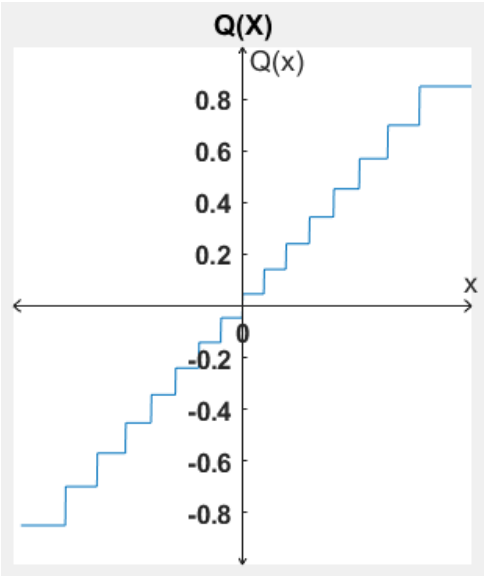

Figure 1.6.2: Non-Uniform Quantization function for N=16

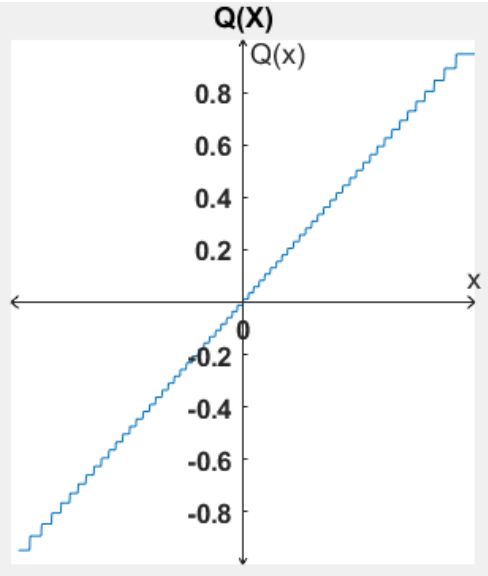

Figure 1.6.3: Non-Uniform Quantization function for N=64

Part-G

We expect the non-uniform quantizers to be strictly better than the uniform quantizers. As expected, SQNR improved across all cases.

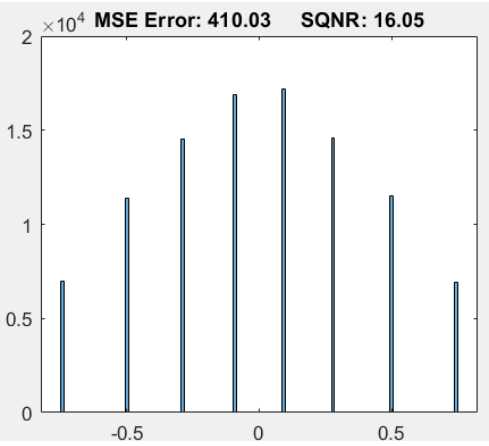

Figure 1.7.1: Non-Uniform Quantization Outputs for N=8

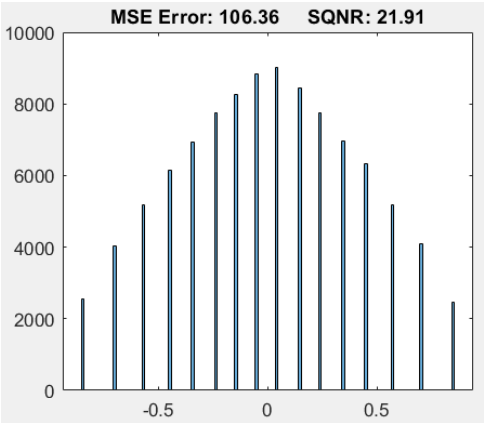

Figure 1.7.2: Non-Uniform Quantization Outputs for N=16

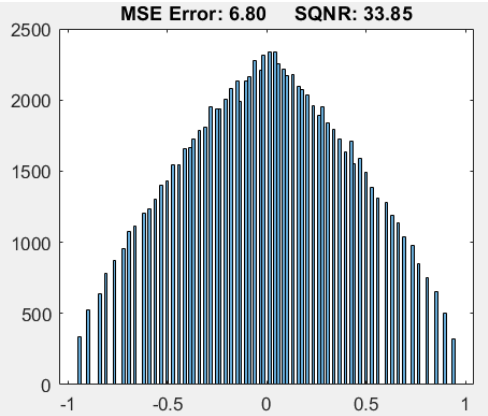

Figure 1.7.3: Non-Uniform Quantization Outputs for N=64

Part-H

We again used the same technique as part-e to calculate the PMF and obtained the following results. The values around 0 became more closely spaced, while the values around 1 became scarcer. This is because values around 0 are more probable and values around 1 are less probable. This spacing is as we expected and minimises the error.

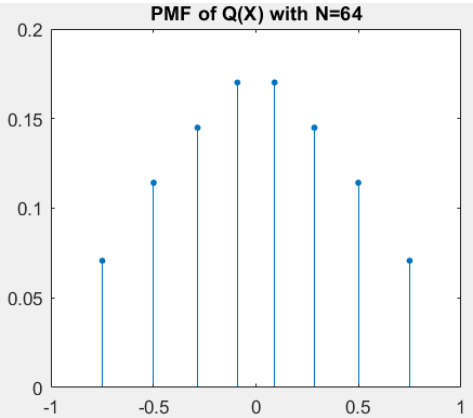

Figure 1.8.1: Non-Uniform Quantization PMF for N=8

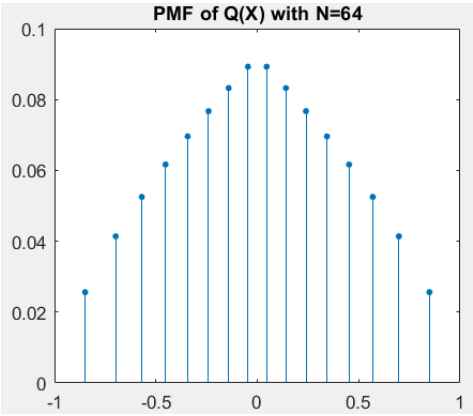

Figure 1.8.2: Non-Uniform Quantization PMF for N=16

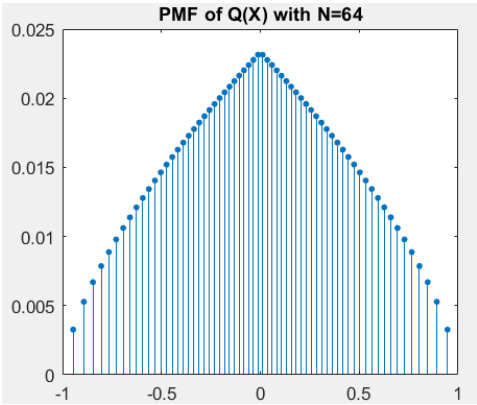

Figure 1.8.3: Non-Uniform Quantization PMF for N=64

Question 2 (foliage.tif) [3]

Part-A

We can plot the histogram of pixel values and also obtain its 4th-order PDF as follows:

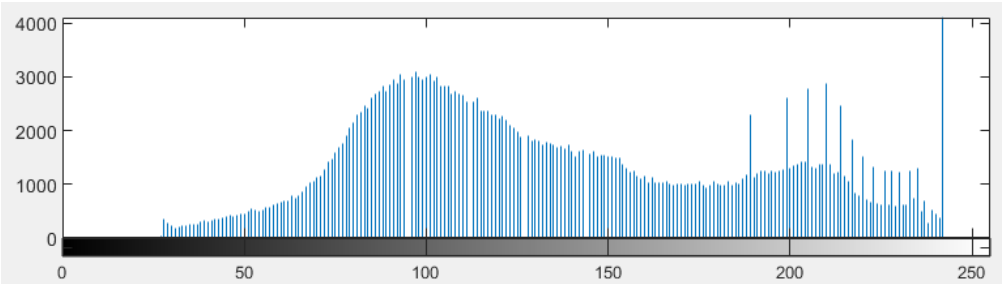

Figure 2.1.1: Image Pixel Histogram for 'foliage.tif'

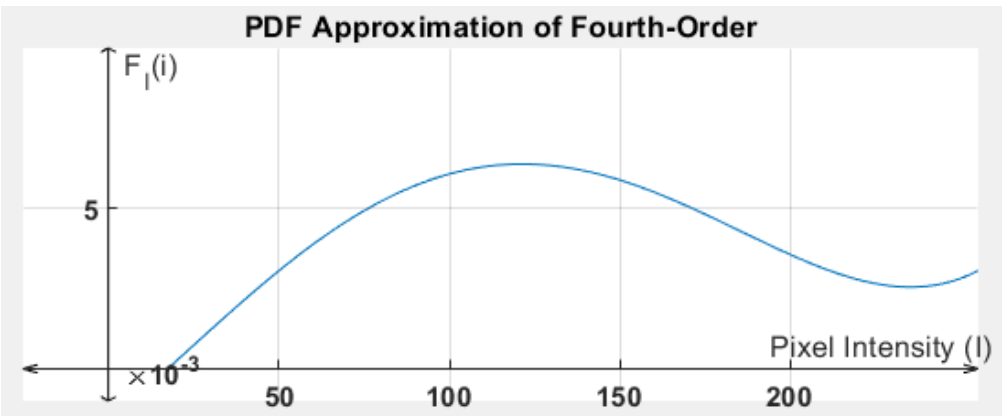

Figure 2.1.2: Image Pixel Histogram and Polynomial Approximation for 'foliage.tif'

Part-B

We can see the image saturation increases for smaller N. Larger N are smoother. We expect an SQNR increase of 6.02 from the increase of 1 bit each step [1].

Figure 2.2.1.1: Quantized Image for N=2

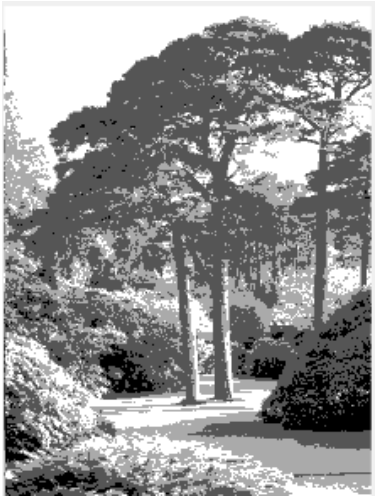

Figure 2.2.1.2: Quantized Image for N=4

Figure 2.2.1.3: Quantized Image for N=8

Figure 2.2.1.4: Quantized Image for N=16

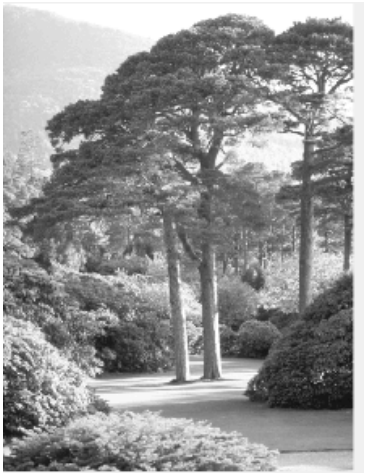

Figure 2.2.1.5: Quantized Image for N=32

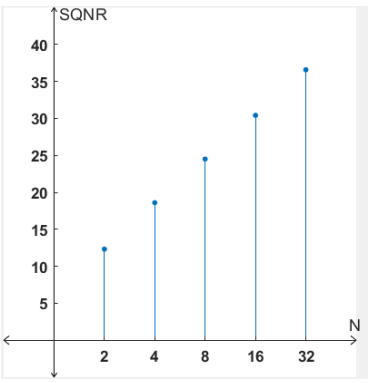

Figure 2.2.2: SQNR as a function of N

	N=2	N=4	N=8	N=16	N=32
MSE $\times 10^8$	3.9904	0.9403	0.2422	0.0620	0.015
SQNR (dB)	12.3261	18.6037	24.4952	30.4102	36.5726

These MSE values are for 10^8 quantized pixels.

Table 1: MSE Error and SQNR Values for Different N Values

Part-C

The signal itself is mainly composed of lower frequencies, and quantization introduces discontinuous jumps, which generate high-frequency noise. The SNR of the noisy FFT signals is correlated with the SQNR of the spatial domain signal. (Note that the fft2s are fftshifted! That means the lower frequency is in the center.)

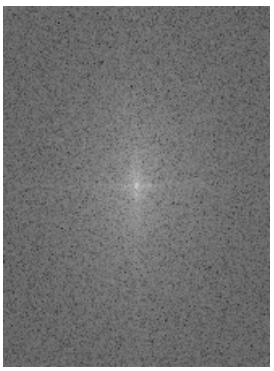

Figure 2.3.1: Quantized Image's FFT for N=2

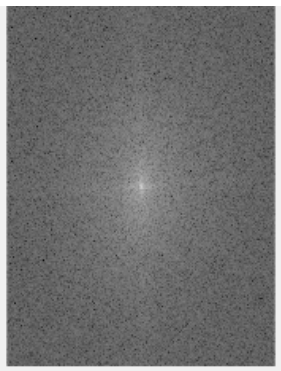

Figure 2.3.2: Quantized Image's FFT for N=4

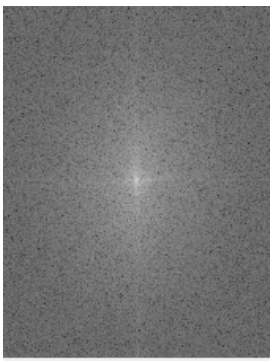

Figure 2.3.3: Quantized Image's FFT for N=8

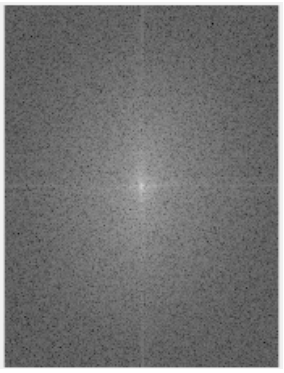

Figure 2.3.4: Quantized Image's FFT for N=16

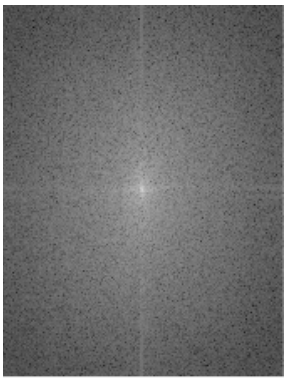

Figure 2.3.5: Quantized Image's FFT for N=32

Part-D

When we apply the operation:

$$H(u,v) = \begin{cases} 1, & |u| < 0.1 \times W, |v| < 0.1 \times H, \\ 0, & \text{otherwise} \end{cases}$$

This acts as an ideal low-pass filter, reducing all high-frequency components of the original image and consequently blurring it. This reduces the image's sharpness. Along with quantization, we have removed both color depth and sharp edges from the image. The image that is LPFd and the image that is quantized and LPFd look almost the same. This is because LPF removes most of the quantization noise, so it does not matter if we quantize the image or not. We have established that quantization noise mostly affects high frequencies.

Figure 2.4.1: Original Image

Figure 2.4.2: LPFd and Quantized Image

Figure 2.4.3: LPFd Image

Question 2 (testimage.tif) [4]

In this section, the results are presented without explanation, as the explanations are identical to those in the previous section. Our observations with this image are in line with our previous findings. This image is nice to use in this context since it features sharp edges and various shades of gray, which will allow us to see the effects clearly.

Part-A

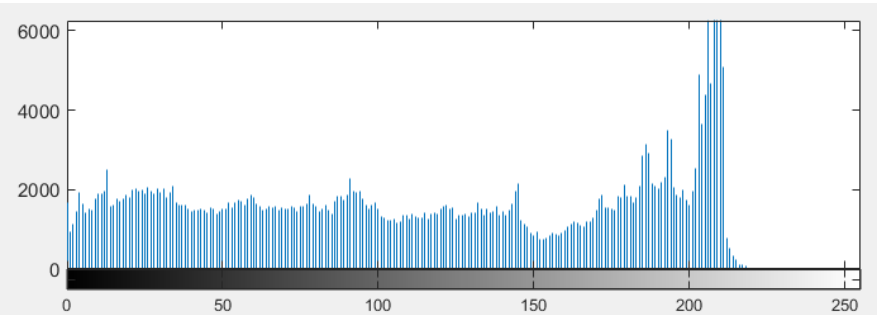

Figure 3.1.1: Image Pixel Histogram for 'testimage.tif'

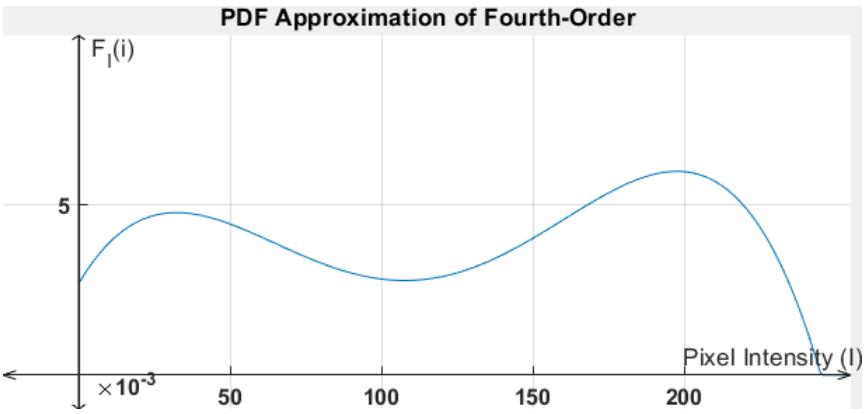

Figure 3.1.2: Image Pixel Histogram and Polynomial Approximation for 'testimage.tif'

Part-B

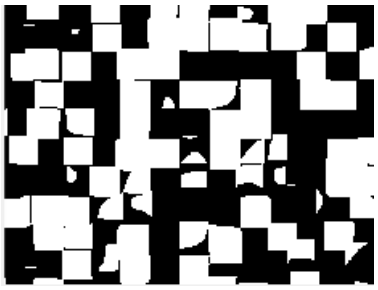

Figure 3.2.1.1: Quantized Image for N=2

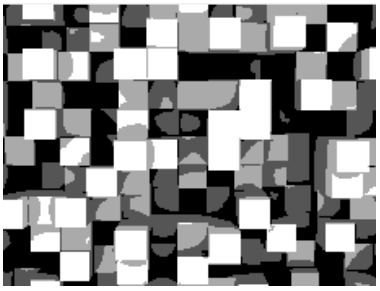

Figure 3.2.1.2: Quantized Image for N=4

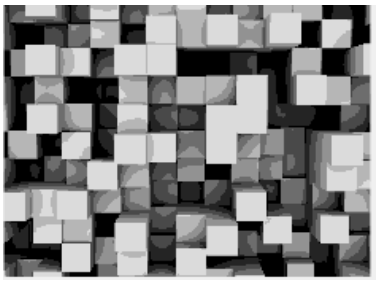

Figure 3.2.1.3: Quantized Image for N=8

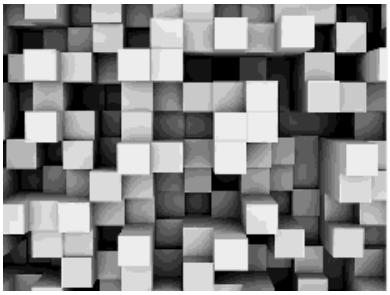

Figure 3.2.1.4: Quantized Image for N=16

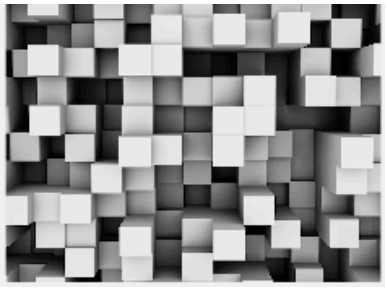

Figure 3.2.1.5: Quantized Image for N=32

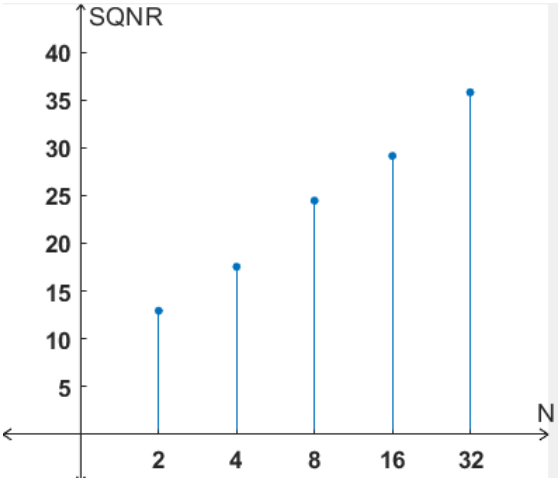

Figure 3.2.2: SQNR as a function of N

	N=2	N=4	N=8	N=16	N=32
MSE $\times 10^8$	4.1622	1.4418	0.2935	0.0999	0.0216
SQNR (dB)	12.9449	17.5492	24.4627	29.1442	35.7953

These MSE values are for 10^8 quantized pixels.

Table 2: MSE Error and SQNR Values for Different N Values

Part-C

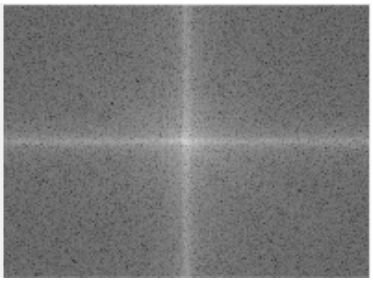

Figure 3.3.1: Quantized Image's FFT for N=2

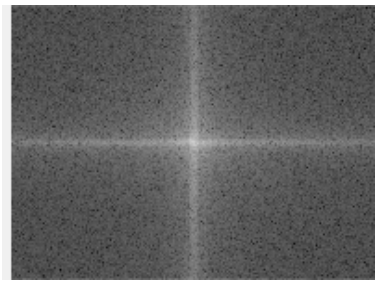

Figure 3.3.2: Quantized Image's FFT for N=4

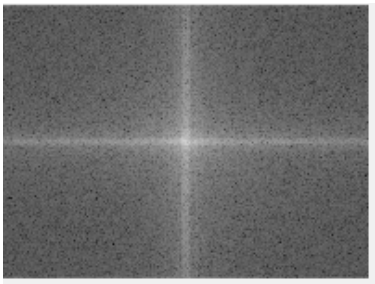

Figure 3.3.3: Quantized Image's FFT for N=8

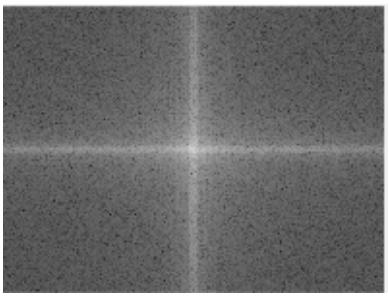

Figure 3.3.4: Quantized Image's FFT for N=16

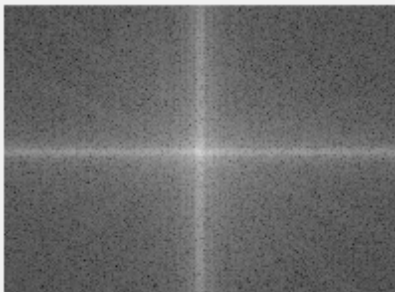

Figure 3.3.5: Quantized Image's FFT for N=64

Part-D

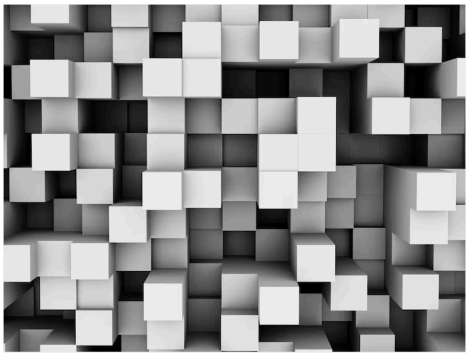

Figure 3.4.1: Original Image

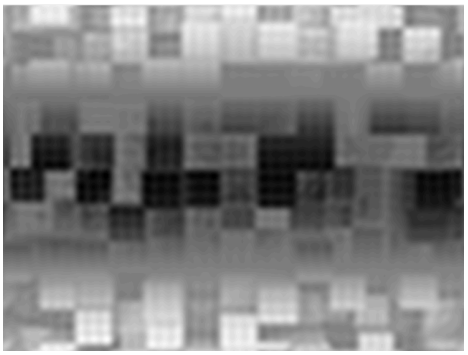

Figure 3.4.2: LPFd and Quantized Image

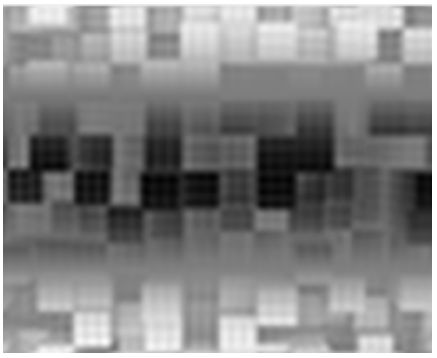

Figure 3.4.3: LPFd Image

Conclusion

In conclusion, in this computational assignment, we have implemented both uniform and non-uniform quantizers for 100,000 random variable outputs corresponding to a given PDF. We calculated the non-uniform quantizer using the Lloyd-Max algorithm and saw a strict increase in SQNR across all cases. We then computed the PMFs for the quantizer outputs.

For the second question, we estimated the PDFs of two images' pixel values. We then quantized the image and investigated the quantization effects both in the spatial and Fourier domains. We saw that quantization increased the saturation of the image in the spatial domain and added noise in the Fourier domain. We then applied an LPF to the quantized image and the original image and found that an LPF introduced blur to the image. We also observed that the LPFd and quantized image were almost identical to the original LPFd image, and we concluded that this is because the LPF removes most of the quantization noise.

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